

1.6 Rules of Inference - 1

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Understanding basic proof structure.

Definition: An argument in propositional logic is a sequence of propositions. Of these propositions, the final one is the conclusion and all others are the premises.

Definition: An argument form is a sequence of compound propositions with propositional variables.

Arguments and argument forms are valid if when the premises are true, then the conclusion is true.

P_1 = "I have bread."
 P_2 = "I put P.B. on bread."
 P_3 = "I put jelly on bread."
 P_4 = "I assemble bread with P.B.
and jelly into a sandwich."
 P_5 = "I have a P.B.J sandwich." → Conclusion

Premises } Argument

$(P_1 \wedge P_2 \wedge P_3 \wedge P_4) \rightarrow P_5$ → Argument form
↓
Premises Conclusion

To show validity of an argument, it is easier to show validity of its argument form.

We still don't want to use truth tables if we can help it. Instead we will discuss validity of simple argument forms which are helpful in analyzing more complex argument forms. The validity of these simple forms are called the

rules of inference

1) Law of Detachment

Also called "modus ponens".

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

p	} Premises
$p \rightarrow q$	
$\therefore q$	> Conclusion

$\therefore$ = Therefore

"If p is true and the implication is true, then q must be true."

2) Modus Tollens

$$(\sim q \wedge (p \rightarrow q)) \rightarrow \sim p$$

$\sim q$
$p \rightarrow q$
$\therefore \sim p$

"If q is false but the implication is true, it must be that p is false."

3) Hypothetical Syllogism

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

$p \rightarrow q$
$q \rightarrow r$
$\therefore p \rightarrow r$

"If $p \rightarrow q$ is true and $q \rightarrow r$ is true, it must be that $p \rightarrow r$ is also true."

4) Disjunctive Syllogism

$$((p \vee q) \wedge \sim p) \rightarrow q$$

$p \vee q$
$\sim p$
$\therefore q$

"If the weather is sunny or cloudy but it is not sunny then it is cloudy."

5) Addition

$$p \rightarrow (p \vee q)$$

p
$\therefore p \vee q$

"If it is sunny, then it is sunny or it is raining."

6) Simplification

$$(p \wedge q) \rightarrow p$$

$p \wedge q$
$\therefore p$

$$\begin{array}{|l} p \wedge q \\ \hline \therefore p \end{array}$$

7) Conjunction

$$(p) \wedge (q) \rightarrow (p \wedge q)$$

$$\begin{array}{|l} p \\ q \\ \hline \therefore p \wedge q \end{array}$$

"If it is sunny and it is raining, then it is both sunny and raining."

8) Resolution

$$(p \vee q) \wedge (\sim p \vee r) \rightarrow (q \vee r)$$

$$\begin{array}{|l} p \vee q \\ \sim p \vee r \\ \hline \therefore q \vee r \end{array}$$

Note: An argument (and its argument form) are valid if the argument form is a tautology.

Verify Modus Ponens is a valid argument.

To be valid, $(p \wedge (p \Rightarrow q)) \Rightarrow q$ must be a tautology (never false).

If $(p \wedge (p \Rightarrow q)) \Rightarrow q$ were false,

then $p \wedge (p \Rightarrow q) \equiv T$ and $q \equiv F$.

If $p \wedge (p \Rightarrow q) \equiv T$ then

$p \equiv T$ and $p \Rightarrow q \equiv T$.

We have $p = T, q = F$ so $p \Rightarrow q$

is $T \Rightarrow F \equiv F$, not T .

Then no solutions exist, so the argument form is never false i.e. a tautology

Then no solutions exist, so the argument form is never false i.e. a tautology thus a valid argument.

Consider the following arguments. What is its argument form and what rule of inference is being used?

"It is either hotter than 100 degrees today or the pollution is dangerous. It is 75 degrees today. Therefore, the pollution is dangerous."

$$((p \vee q) \wedge \sim p) \rightarrow q \text{ Disj. Syllogism}$$

"It is either hotter than 100 degrees today or the pollution is dangerous. It is 98 degrees today or the humidity is high. Therefore, the pollution is dangerous or the humidity is high."

$$((p \vee q) \wedge (\sim p \vee r)) \rightarrow (q \vee r) \text{ Resolution}$$

What valid conclusions can be drawn?

$$p = \begin{cases} \text{"If I take the day off, it} \\ \text{either rains or snows."} \end{cases} \begin{cases} u = \text{I take the day off.} \\ v = \text{It rains.} & w = \text{It snows.} \\ p = u \rightarrow (r \vee w) \end{cases}$$

$$\begin{cases} q = \text{"I took Tuesday off or I} \\ \text{took Thursday off."} \\ r = \text{"It was sunny on Tuesday."} \\ s = \text{"It did not snow on Thursday."} \end{cases} \begin{cases} x = \text{It is Tuesday.} \\ y = \text{It is Thursday.} \end{cases}$$

$$(p \wedge r) \rightarrow \text{"I did not take off on Tuesday."}$$

$$p \wedge r \equiv \underbrace{(u \rightarrow (r \vee w))}_1 \wedge \underbrace{(\sim (r \vee w))}_2 \wedge \underbrace{x}_3$$

$$\text{regroup} \equiv \left(\underbrace{\left((u \rightarrow (r \vee w)) \wedge (\sim (r \vee w)) \right)}_{\downarrow} \right) \wedge x$$

Modus Tollens says the valid conclusion is $\sim u$
 - "I did not take off on Tuesday."

valid conclusion is $\sim u$
or "I did not take off."

$\Rightarrow (\sim u) \wedge x$ is the complete
conclusion by Addition

$\Rightarrow$ "I did not take off on Tuesday."

• $(q \wedge (p \vee r)) \rightarrow$ "I took off on Thursday."

$q =$ "I took Tuesday off or I took Thursday off."

$q = (u \wedge x) \vee (u \wedge y)$

$p \wedge r \rightarrow \sim u \wedge x$ (previous conclusion drawn)

$\Rightarrow (q \wedge (p \vee r)) \equiv ((u \wedge x) \vee (u \wedge y)) \wedge (p \vee r)$

$\Rightarrow \underbrace{((u \wedge x) \vee (u \wedge y))}_{\text{Assumed True}} \wedge \underbrace{(\sim u \wedge x)}_{\text{Assumed True}}$

Disjunctive Syllogism gives that

it must be the case that the

conclusion is $\sim u \wedge y$.

$\rightarrow \sim u \wedge y$ "I took off on Thursday."

Is the following argument valid?

"Raining days make gardens grow."

"Gardens don't grow if it is not hot."

"It always rains on a day that is not hot."

"Therefore, if it is not hot, then it is hot."